# Herasat Portal — Database Migration Runbook

PostgreSQL Complete Dump, Network Transfer & Clean Restoration Protocol

HMD IT HERASAT PRODUCTION GUIDE

1 Dump (Source) → 2 Transfer File → 3 Copy Uploads → 4 Restore (Dest) → 5 Verify Records

**CRITICAL SECURITY MANDATE — AES-256 ENCRYPTION KEY:**
The environment variable `HERASAT_AES_KEY` must be **100% identical** on both the source and destination servers. Encrypted columns (citizen descriptions, messages, replies, and confidential notes) are encrypted at rest with AES-GCM. If the destination server uses a different key, existing reports will permanently fail decryption.

## 1 Database Credentials & Environment Profile

| Parameter | Standard Configuration | Notes |
|---|---|---|
| **Database Name** | `herasat_db` | Core database containing citizen reports and user accounts |
| **Database User** | `herasat_user` | Dedicated service account with ownership of the tables |
| **Default Password** | `herasat1405@` | Supplied when prompted by pg_dump / pg_restore |
| **PostgreSQL Path (Win)** | `C:\Program Files\PostgreSQL\18\bin\` | Contains `pg_dump.exe`, `pg_restore.exe`, `psql.exe` |
| **Uploaded Attachments** | `./uploads/` folder in project root | Stores uploaded PDF, JPG, PNG evidence files |

## 2 Source Server: Creating the Backup Dump

We use PostgreSQL custom compressed archive format (`-F c`). It creates a compressed binary file that allows parallel restoration, automatic dependency ordering, and clean table rebuilding.

POWERSHELL (SOURCE SERVER)

```
& "C:\Program Files\PostgreSQL\18\bin\pg_dump.exe" -h 127.0.0.1 -U herasat_user -d herasat_db -F c -b -v -f "C:\Users\Novin\herasat_db.dump"
```

COMMAND PROMPT (CMD.EXE)

```
"C:\Program Files\PostgreSQL\18\bin\pg_dump.exe" -h 127.0.0.1 -U herasat_user -d herasat_db -F c -b -v -f "C:\Users\Novin\herasat_db.dump"
```

**Dump File Location:** The output file will be saved at `C:\Users\Novin\herasat_db.dump` (~43 KB). You can locate and copy this file directly via Windows Explorer or USB / SCP.

## 3 Transferring Files to Destination Server

### A. Database Archive File

- Copy `herasat_db.dump` from source PC.
- Place it at `C:\herasat_db.dump` on the destination server for simple command access.

### B. Uploaded Evidence Files (Must Copy)

- Copy the entire `uploads\` folder from `herasat_backend\`.
- Paste into destination `herasat_backend\uploads\`.
- Ensures citizens' attached evidence files remain intact.

## 4 Destination Server: Restoring the Database

Run the restore utility on the destination Windows Server. The `--clean --if-exists` flags safely drop existing tables before importing, ensuring no duplicate primary key collisions.

IN COMMAND PROMPT (CMD.EXE) AS ADMINISTRATOR:

```
"C:\Program Files\PostgreSQL\18\bin\pg_restore.exe" -h 127.0.0.1 -U herasat_user -d herasat_db --clean --if-exists -v "C:\herasat_db.dump"
```

IN POWERSHELL AS ADMINISTRATOR:

```
& "C:\Program Files\PostgreSQL\18\bin\pg_restore.exe" -h 127.0.0.1 -U herasat_user -d herasat_db --clean --if-exists -v "C:\herasat_db.dump"
```

* When prompted for password, enter: `herasat1405@`

## 5 Common Errors & Troubleshooting

**Error: '& was unexpected at this time'**

**Cause:** You pasted a PowerShell command starting with & into a standard Windows Command Prompt (`cmd.exe`).
**Fix:** Remove the leading & symbol in `cmd.exe`, or type `powershell` first before running.

**Error: 'pg_dump' or 'pg_restore' is not recognized as an internal or external command**

**Cause:** PostgreSQL's `bin` folder is not in Windows system PATH.
**Fix:** Use the full path with quotation marks: `"C:\Program Files\PostgreSQL\18\bin\pg_restore.exe"`.

**Error: Path not found / Version mismatch**

**Cause:** Destination server might have PostgreSQL 16, 17, or custom install folder instead of 18.
**Fix:** Check `C:\Program Files\PostgreSQL\` and replace `18` with the installed version number.

## 6 Post-Restoration Verification

Run this SQL verification command on destination to confirm all reports and users are loaded:

COMMAND PROMPT

```
"C:\Program Files\PostgreSQL\18\bin\psql.exe" -h 127.0.0.1 -U herasat_user -d herasat_db -c "SELECT count(*) AS total_reports FROM reports; SELECT count(*) AS total_users FROM admin_users;"
```

## 7 Starting the Backend Server on Destination

POWERSHELL (DESTINATION PROJECT FOLDER)

```
$env:HERASAT_AES_KEY="HERASAT_SECURE_KEY_32_CHARS_LONG"
$env:HERASAT_DB_CONN="host=127.0.0.1 port=5432 user=herasat_user password=herasat1405@ dbname=herasat_db sslmode=disable"
go run .
```